

Python-based computational platform to automate seismic design, nonlinear structural model construction and analysis of steel moment resisting frames

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ABSTRACT

An end-to-end computational platform is presented, which automates seismic design, nonlinear structural model construction, and response simulation (static and dynamic) of steel moment resisting frames. A modular framework is adopted along with the object-oriented programming paradigm to ensure the adaptability of the platform. The seismic design module iteratively generates code-conforming section sizes and detailing for beams, columns, and beam-column connections based on the relevant input design variables including the building configuration (e.g., the number of stories, the number of lateral-force resisting systems, and the building dimensions), loads (e.g., dead and live loads on each floor), and site conditions (mapped spectral acceleration parameters). The nonlinear model construction and analysis module takes the design results as input and produces structural models that capture flexural strength and stiffness deterioration in the frame beam-column elements, and performs pushover and response history analyses. Illustrative examples are presented to demonstrate the reliability, accuracy, and efficiency of the platform, which significantly reduces the time and effort involved in producing iterative structural designs and conducting nonlinear analyses, both of which are necessary for performance-based seismic design. Additionally, the platform can be used to create an extensive database of archetype steel moment frame buildings towards the development of analytics-driven design methods.

1. Introduction

Steel moment resisting frames (SMRFs) are often used as a part of the lateral force-resisting systems (LFRS's) in buildings designed to resist earthquakes. Steel special moment frames represent one of the few LFRS options that is permitted without restrictions in ASCE 7–16 [1] for buildings exceeding 160 ft (48.77 m) in regions of high seismicity. It is well-known that SMRFs are able to provide significant inelastic deformation capacity through flexural yielding at the beam ends and limited yielding in panel zones, which enables ductile response in moderate-to-severe earthquakes. Another advantage of SRMFs is that they do not require structural walls or diagonal braces and therefore offer an unobstructed line of sight, which provides flexibility in architectural design. A recent study by Hamburger and Malley [2] showed that SMRFs typically impose smaller forces on foundations compared other structural systems, resulting in more economical sub-structure systems. Because of these advantages, SMRFs have been widely used in industrial plants, low- and mid-rise residential and commercial buildings, and some tall buildings as part of a dual LFRS.

There exists an abundance of experimental and numerical studies on developing reliable modeling techniques for SMRFs [3–5], quantifying the influence of the gravity system on collapse risk [6,7], improving the seismic performance through the use of novel devices [8–11] and quantifying earthquake-induced socioeconomic impacts, such as economic losses, fatalities, and downtime [12–14]. Because most of these studies were performed on one or a few prototype buildings designed to comply with older or modern building code provisions, their findings cannot be generalized. Ideally, a large number of building cases should be used to systematically capture variations in key structural characteristics (e.g., number of stories, number of bays, bay widths, story heights and magnitude of dead loads).

To address the challenge of creating a generalizable design space, the concept of archetype buildings has been proposed. Buildings with similar geometric configuration and/or structural properties are grouped into representative archetypes. The archetype concept bridges the gap between performance predictions for a specific building and the generalized predictions for a full class of structures [15]. A typical starting point for archetype studies in the domain of seismic design,

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analysis and performance-based assessment is to create a design space by identifying the variables that affect seismic performance and establishing the bounds for each one. Next, each representative archetype building is designed to comply with the relevant building code(s) and standard(s). Subsequently, nonlinear structural models are constructed using an appropriate computational platform and seismic responses are obtained through the relevant analysis procedures. The final steps are used to synthesize the results and develop general conclusions regarding the seismic performance for an entire class of buildings.

The archetype concept has been extensively used in recent years to investigate the seismic performance of building structures. Some of the earliest work in this area utilized the archetype approach to assess the collapse risk of both modern (ductile) and older (non-ductile) reinforced concrete (RC) special moment frame buildings [16–18]. The Applied Technology Council (ATC) developed a systematic approach to assess seismic design provisions for building LFRS's based on archetype buildings/models [19]. Soon after, the archetype concept was adopted as part of the FEMA P695 [20] guidelines, which provides a comprehensive methodology for quantifying the collapse vulnerability of different LFRS's. Since then, several other studies utilizing archetypal buildings have been published (e.g., [7,21,22]). Given the growing significance and popularity of the archetype concept in performance-based earthquake engineering, there is a need to develop a set of computational tools and processes to automate seismic design, nonlinear structural model construction and seismic response analyses (static and dynamic).

The seismic designs reported in the relevant literature [3,15,21,23–25] and engineering practice are all performed using an iterative process that is based on extensive interactions between the engineers and the design software. It typically starts by assigning preliminary member sizes based on engineering judgement and established rule of thumbs. The design is then evaluated with the help of the software to determine whether it complies with the relevant building codes and standards. Based on the evaluation results, the design is revisited accordingly. This process is repeated until the design satisfies all the relevant requirements and is not excessively conservative. The major drawback of this process is the extensive involvement of human effort in repeatedly adjusting designs. While several studies [26–28] in structural engineering introduced the concept of “computer-automated design”, they are all based on old versions of the building code and no computation platforms were produced to facilitate the design. Consequently, the designs are still performed in an iterative manner with extensive human effort. To significantly reduce the engineers' effort during the iterative process and to explore the possibility of seismic design automation, a computational platform with appropriate algorithms, data structures, and programing paradigm is developed.

This study introduces a Python-based platform that is able to automate the seismic design, nonlinear structural model generation and response analyses for SMRFs. The Automated Seismic Design and Analysis (AutoSDA) platform is developed using the object-oriented programing paradigm and has the two modules shown in Fig. 1. This first AutoSDA module takes in building geometry and load information as input and generates code-conforming designs (e.g., SMRF member sizes and beam-column joint details). The second module takes in the design information generated by the first module as input and constructs two dimensional (2D) nonlinear structural models. It further automates the process of conducting nonlinear static and dynamic analyses and post-processing the results. As shown in Fig. 1, the linear static (during the design process) and nonlinear analyses (of the final design) are performed in the Open System for Earthquake Engineering Simulation (OpenSees) [29]. All other features have been implemented using original Python scripts.

There are several intellectual contributions that are embedded in the AutoSDA platform. First, the object-oriented programing paradigm is used to automate the SMRF seismic designs, which has the potential to revolutionize the design process. While the current version of the

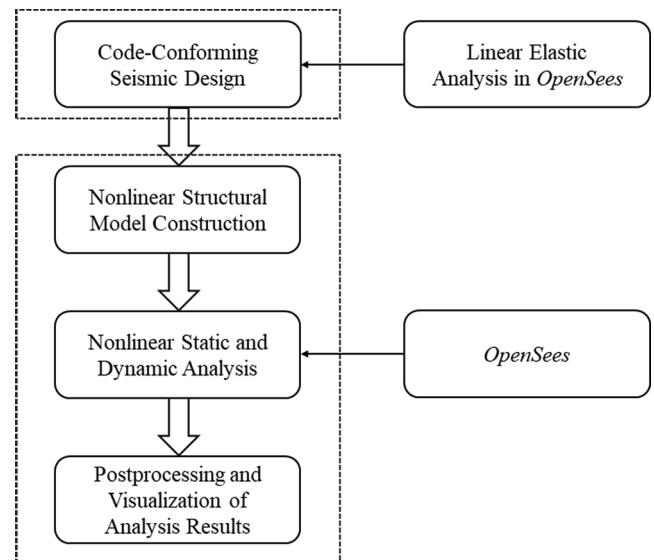


Fig. 1. Overview of the main AutoSDA platform modules.

AutoSDA platform is specific to SMRF's, it is developed and presented in a generalized manner such that it can be extended to other LFRS's. Also, several of the sub-algorithms (e.g., adjusting member sizes to meet drift requirements, evaluating the design requirements for beams, columns, and connections, checking for constructability while maintaining design constraints) can be adapted to other design contexts. Second, the AutoSDA platform enables users to conveniently bridge the gap between design variables and seismic design and performance outcomes. Consequently, it could be used to systematically investigate the influence of different design variables on the design outcome and associated seismic performance (e.g. effect of lateral strength and collapse resistance). Additionally, the AutoSDA platform drastically reduces the time and effort involved in generating code-conforming designs and constructing nonlinear structural models. As such, it could be used to generate a dataset of code-conforming SMRFs. One example of such a dataset generated by AutoSDA can be found in [30]. It includes 621 SMRFs that have been designed in accordance with modern codes and standards, the corresponding nonlinear structural models, and seismic responses. To the best of the authors' knowledge, this is the largest among all currently publicly available seismic design and response datasets. Because of this, it can be used to evaluate existing simplified seismic demand estimation methods and develop new data-driven models. Last, the AutoSDA platform enables the possibility of using computer-automated (rather than computer-aided) design in the area of structural seismic design. The inclusion of the details of how the platform is developed and the open source codes have the potential to help researchers advance truly automated structural designs. The remainder of the paper begins by introducing the design criteria and modeling strategies for SMRFs. Then, the AutoSDA structure and development details are presented. Illustrative examples are provided to demonstrate the reliability and efficiency of the platform. Finally, the limitations and possible extensions of the AutoSDA platform are discussed.

2. Seismic design and nonlinear structural modeling of SMRFs

2.1. Overview of design criteria

The core aspect of seismic design for a SMRF is to determine the sizes and detailing for three basic components: beams, columns, and beam-column connections. Three analysis procedures, specifically equivalent lateral force (ELF), response spectrum analysis (RSA), and response history analysis (RHA), are available to determine the design forces and deformations. The ELF method is adopted in the current

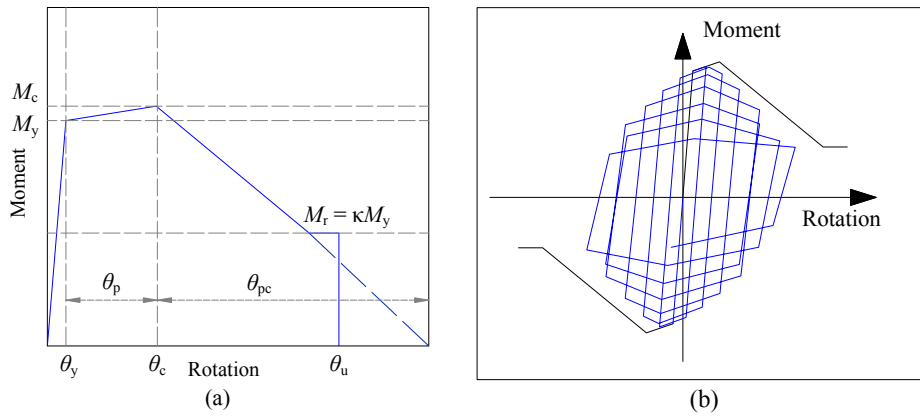


Fig. 2. Modified IMK material model: (a) monotonic backbone curve and (b) cyclic response.

version of the computational platform with the understanding that other alternative analysis methods (RSA and RHA) can be added in the future. The following main steps are used to design a SMRF using the ELF analysis method:

Step 1: Seismic story forces are determined based on the approximate fundamental period computed using Equation 12.8-7 of ASCE 7-16 [1]. Preliminary sizes for beams and columns are determined, typically based on engineering judgement and established rules of thumb. Accidental torsion is considered in accordance with Section 12.8.4.2 of ASCE 7-16 by assuming that the center of mass deviates from the geometric centroid by a distance that is 5% of the diaphragm dimension perpendicular to the direction of the applied forces.

Step 2: An elastic model of the SMRF is constructed and analyzed for a combination of gravity and lateral seismic story forces. Structural deformations and member forces are obtained from the elastic analysis.

Step 3: The story drifts obtained from Step 2 are adjusted using the deflection amplification factor (as specified in Table 12.1-1 of ASCE 7-16) and compared with the drift limit. The drift limit for SMRFs with seismic design categories D, E, or F is taken as Δ_a/ρ , where Δ_a is the limit specific by Table 12.12-1 of ASCE 7-16 [1] and ρ is the redundancy factor determined by Section 12.3.4 of ASCE 7-16 [1]. For other seismic design categories, the drift limit is Δ_a . It is worth noting that the story drifts here should include the diminished beam stiffness if reduced beam section (RBS) connections are adopted. If the drifts are found to be greater than the limit, the member sizes are increased, and Steps 1 to 3 are repeated. As prescribed in Section 12.8.7 of ASCE 7-16 [1], frame stability should also be checked at this stage unless P- Δ effects have been explicitly incorporated in the elastic analysis.

Step 4: Column and beam sections are checked to ensure that they satisfy the “highly ductile member” requirement, which requires the width-to-thickness ratios of flanges and webs to be less than the limit specified in Table D1.1 of ANSI/AISC 341-16 [31]. The ability of columns to resist axial loads, shear, bending moments, and axial-flexural demands is evaluated using the equations specified in ANSI/AISC 360 [32] Chapters E3, G2, F2, and H1, respectively. All of the demands are determined using the load combinations provided in Sections 2.3 and 2.4 of ASCE 7-16 [1]. Similarly, beams are checked to ensure that they have adequate strength to resist shear and flexural demands. Additionally, the need for lateral bracing in beams is evaluated to avoid lateral-torsional buckling.

Step 5: A commonly used prequalified connection, the reduced beam section (RBS), is adopted in the current version of the AutoSDA platform with the understanding that other types of connections (e.g., semi-rigid) can be incorporated in the future. The shear and flexural capacities of the beam-column connections must be greater than the probable maximum shears and moments. According to AISC 341 Chapter E3, the column-beam moment ratio shall be greater than 1.0 to ensure that the strong-column-weak-beam (SCWB) design criterion is met. For a

single-story building or connections at the roof of a multi-story building, the SCWB requirement is exempted. The need for doubler plates and continuity plates is also evaluated in this step.

Step 6: At this stage, a design that satisfies all the requirements specified in the relevant building code and standards is obtained. However, the design sizes might not be the most suitable for construction purposes. For example, the same member size is typically used in every two or three adjacent stories. Additionally, lower stories generally have deeper columns than upper stories to accommodate splice connections and the beams at the lower floor levels are typically deeper and stronger than the ones above.

Further details regarding the seismic design of SMRFs can be found in the relevant building code and design standards (e.g., [1,31,32]).

2.2. Nonlinear modeling of SMRFs

Key to the development of nonlinear structural models for SMRF's is the numerical representation of beams, columns, and beam-column connections. For nonlinear analyses ranging from the onset of damage to collapse, the adopted technique should ideally capture the strength and stiffness deterioration of these structural components, which has been shown to significantly influence structural seismic responses. The widely used concentrated plasticity model (CPM) is adopted in the current version of the AutoSDA platform.

Concentrated plastic hinge beam-column elements consist of a linear elastic portion with inelastic hinges at both ends, which are typically represented as zero-length rotational springs. The modified Ibarra-Medina-Krawinkler (IMK) material model is often used in nonlinear hinge elements [33,34]. It has been developed and adapted over the years to simulate the hysteretic behavior of beam-column connections while incorporating both cyclic and in-cycle degradation. As shown in Fig. 2, its monotonic response envelope includes three segments: elastic, hardening, and post-capping. The entire monotonic backbone curve is defined by three strength parameters and three deformation parameters. The yield (M_y), capping (or peak moment) (M_c) and residual moments (M_r) are the strength parameters. The deformation parameters include the yield rotation (θ_y), rotation at peak moment (θ_c), and the rotation at which the strength degrades to zero (θ_u). Three types of cyclic deterioration, including basic strength, post-capping strength, and unloading stiffness, are incorporated by defining six relevant parameters. While the material model requires a total of 24 parameters, past studies [3,4,34,35] have provided empirical equations and qualitative insights on how to determine each parameter. In the OpenSees models constructed by the current version of AutoSDA, the parameters for the beam and column hinges are determined using the empirical equations reported by Lignos and Krawinkler [4], and Lignos et al. [35], respectively. CPM is fairly simple to implement and is computationally efficient, which is partly why it has been widely

embraced. While the implemented version of the CPM cannot capture P-M interaction, past studies have demonstrated that it is reliable enough to estimate structural responses under earthquakes [3,4,36].

Apart from the beam and column components, the possible shear yielding at the SMRF panel zones is also considered. Panel zone yielding usually initiates from the center towards the four corners, which causes parallelogram-shaped deformations. The shear distortion relationship developed by Krawinkler [37] is used to simulate this behavior. The panel zone is modeled using a combination of elastic elements and zero length rotational springs. Eight elastic elements with very high axial and flexural rigidity are used as the boundary elements, which form a parallelogram with a width corresponding to the column depth and a height that is the same as the beam depth. A trilinear rotational spring is placed in one of the four corners to capture the shear distortion. The remaining three corners are modeled as pinned connections. The thickness of the panel zone is taken as the sum of the column web and doubler plate thicknesses.

A leaning column is included to account for P- Δ effects. The leaning column is connected to the frame through a truss element. The hinge for the leaning column is modeled as a zero-length rotational spring with very small rotational stiffness so that it does not add lateral stiffness to the structure. The gravity load on the SMRF is uniformly applied to the beam elements, whereas the load on the part of the gravity system that is not explicitly modeled is applied to the leaning column. The floor mass is uniformly assigned to each node at the same level. A schematic illustration of a five-story, four-bay SMRF model is presented in Fig. 3.

3. Seismic design module

3.1. Overview

An overview of the workflow for the seismic design module is

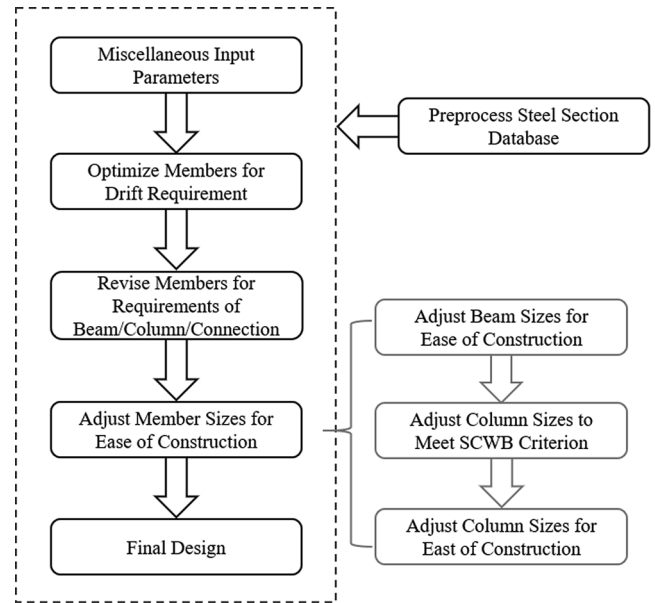


Fig. 4. Overview of the seismic design module.

presented in Fig. 4. A one-time preprocessing of the electronic steel section database is first performed. Then, the relevant seismic design parameters are received as input, beam and column sizes are initialized, and the member-sizes are adjusted to satisfy the drift requirement. Subsequently, the beams, columns, and connections are checked to ensure that they satisfy the relevant strength and detailing requirements and the member sizes are revised as needed. Lastly, the member sizes are adjusted to account for ease of construction and the final

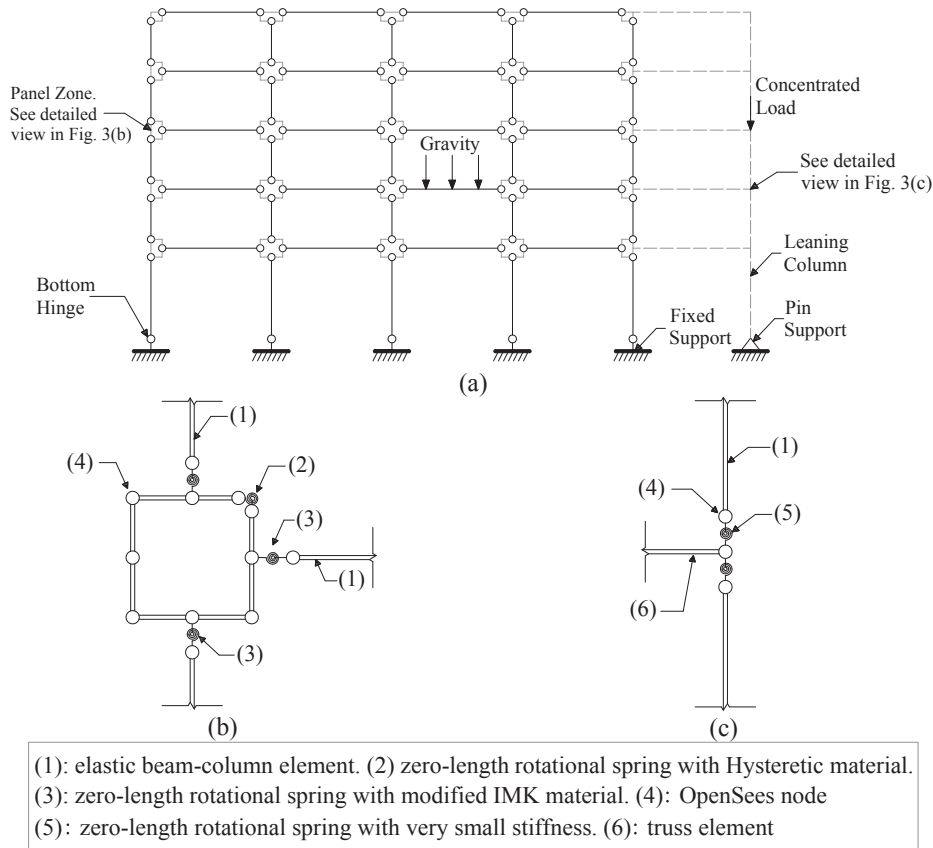


Fig. 3. Nonlinear model for the SMRF: (a) overview of the model, (b) beam-column connection and (c) leaning column joint.

design is generated. The details of each of the main steps in the procedure are presented in the following sections.

3.2. Preprocessing the electronic database of wide flange sections

To facilitate the iterative adoption of SMRF beam and column sizes, the electronic database of wide flange sections provided by the American Institute of Steel Construction (AISC) is pre-processed. The entire database includes all wide flange shapes that are currently manufactured in the industry. However, only those satisfying the high ductility requirement can be used as a SMRF column or beam section. Moreover, the adoption of RBS connection introduces more stringent requirements on the beam and column sections. More specifically, the section depth, weight, and flange thickness of beams must be less than W36, 300 lb/ft (446 kg/m), and 1.75 in. (4.42 cm), respectively. Also, column section depths must be less than W36. Based on these requirements, the original database is filtered to create two new sub-databases of all possible beam and column sections. The section sizes are listed in descending order of the moment of inertia and the plastic section modulus in the column and beam sub-database, respectively. An index beginning from zero and incremented at values of one is attached to each sub-database to denote the order of section sizes (zero represents the strongest section).

3.3. Design automation algorithm

The sub-algorithm used to optimize the member sizes to meet the relevant drift requirement is presented in Fig. 4. Two important coefficients are predefined by the user: the moment of inertia ratio between the exterior and interior columns ($I_{col,ext}/I_{col,int}$) and the plastic section modulus ratio between the beam and interior column ($Z_{bm}/Z_{col,int}$). These two coefficients impose additional constraints on the beam and column section sizes, which will be enforced throughout the entire design process. Based on a review of industry-generated SMRF designs [23], $I_{col,ext}/I_{col,int}$ typically ranges from 0.6 to 0.8. The typical range for $Z_{bm}/Z_{col,int}$ is 0.7 to 0.8 for buildings with less than 10 stories and 0.45 to 0.7 for taller buildings.

As shown in Fig. 5, after taking in the relevant input parameters, the algorithm begins by initializing all the beams and columns in the SMRF with the maximum allowable sizes specified in their respective section

sub-database. Then a linear elastic model of the SMRF is constructed in the Open System for Earthquake Engineering Simulation (OpenSees) [29] platform and subjected to equivalent lateral story forces in accordance with ASCE 7 Section 12.8. Subsequently, the story drifts obtained from the elastic analysis are compared with the allowable limit: 2% at the design basis earthquake (DBE) hazard level. The design module also allows the user to specify a different drift limit based on a desired performance objective. Fig. 5 summarizes the “brute force” approach that is used to target the desired design drift level. First, the story that has the minimum drift is identified and labeled as the “target story”. The size of the interior columns in the “target story” is decreased such that the new section is just one index higher than the previous one in the column section sub-database. Then, the design of the beams and exterior columns is revisited to ensure that $I_{col,ext}/I_{col,int}$ and $Z_{bm}/Z_{col,int}$ are within some acceptable tolerance of the value defined by the user. At this stage, new section sizes are assigned to the members in the “target story” and the elastic analysis is repeated to obtain updated drift demands and member forces. This process is repeated until the maximum story drift exceeds 2%, which is followed by a comparison between the sections in the current and initial designs (which is based on the maximum allowable sections). If these two designs are the same, the implication is that no valid design exists within the member sub-databases. Otherwise, the current member sizes are reduced by one index at a time until the design drift is less than (but closest to) the allowable limit. At this stage, the member sizes have been optimized to meet the drift requirement. Since P-Δ effects are explicitly included in the elastic analysis, a frame stability check is not implemented.

As shown in Fig. 6, after the member sizes are proportioned to meet the drift requirements, component-by-component checks are performed. While SMRF beam and column sizes are typically governed by the drift requirements, strength evaluations are still performed to ensure the robustness of the algorithm. Each column is individually evaluated to ensure that the axial, shear, flexural, and axial-flexural strength requirements specified in ANSI/AISC 360 [32] Chapters E3, G2, F2, and H1, respectively, are satisfied. If a column does not satisfy at least a single strength requirement, its section size is increased such that the new size is one index lower in the column section sub-database. Based on the newly determined column size, the elastic analysis is repeated to ensure that the drift demands are still within the required limit. Meanwhile, the force demands in each member are updated based

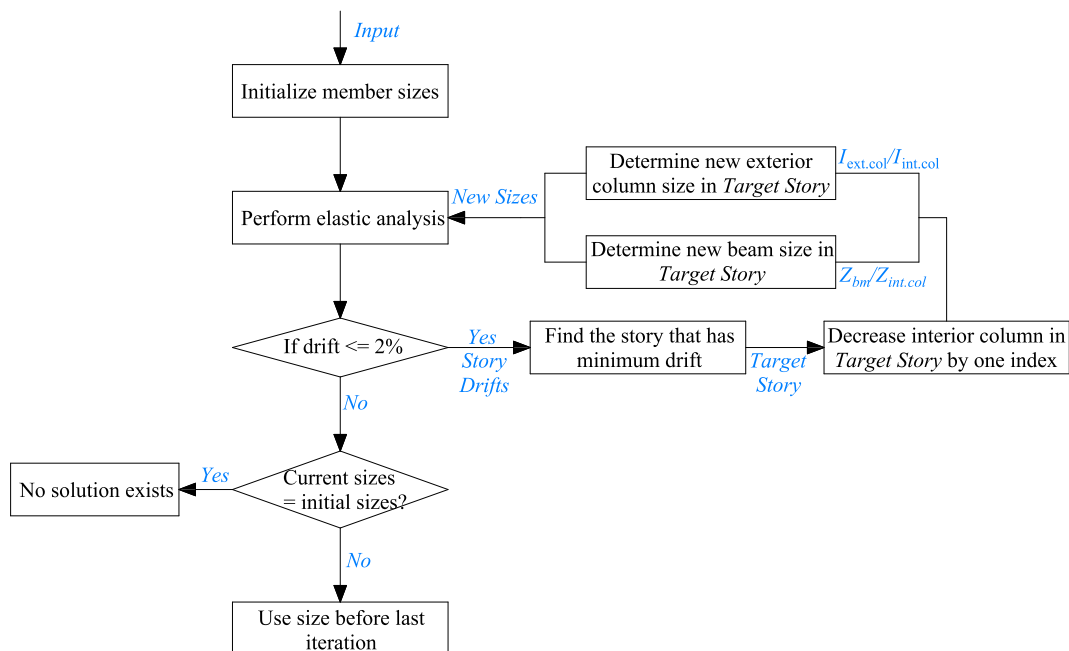


Fig. 5. Overview of sub-algorithm used to achieve the desired target drift demand.

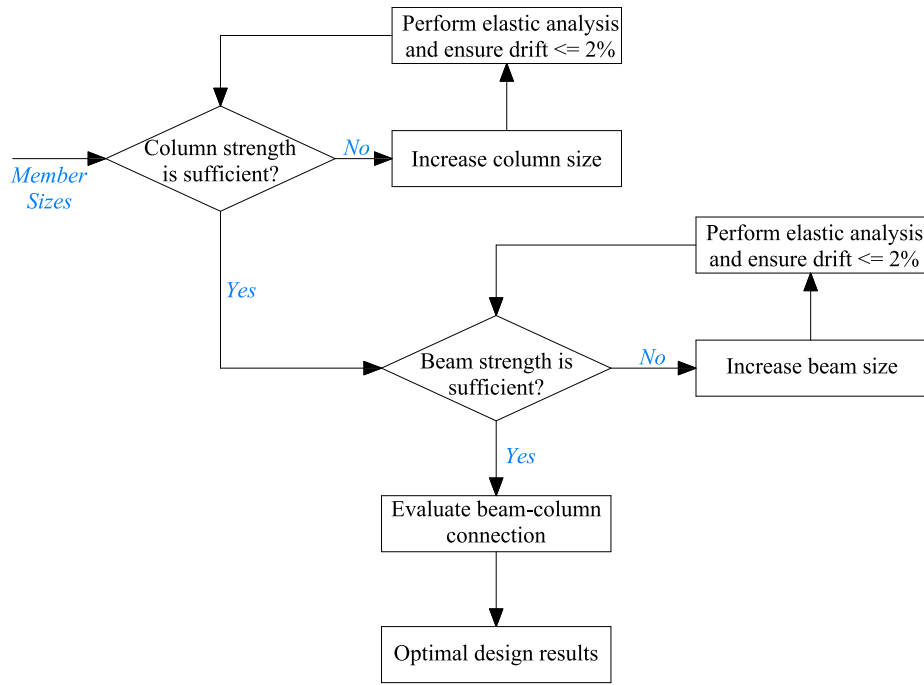


Fig. 6. Overview of sub-algorithm used to check the feasibility of beams, columns and connections.

on the results of this analysis. This procedure is repeated until all columns in the SMRF satisfy the relevant strength requirements. A similar process is implemented for the beams to ensure that they all satisfy the relevant requirements.

The algorithm subsequently evaluates each beam-column connection based on the requirements specified in AISC 358 [38] Chapter 5. First, the beams are checked to ensure that they can resist the demands based on the expected flexural strength at the center of the RBS. The beam-column connections are then checked to ensure that they comply with the SCWB criterion. As shown in Fig. 7, the following parameters are first computed: (1) the required shear strength of the beam (V_u), which is based on the probable moment and shear caused by gravity, (2) the probable maximum moment at the column face (M_f), (3) the shear capacity of the beam (V_n), (4) the plastic moment of the beam based on expected yield stress (M_{pe}), (5) the sum of the moments in the column above and below the joint at the intersection of the beam and column centerlines ($\sum M_{pc}^*$), and (6) the sum of moments in the beams at the intersection of the beam and column centerlines ($\sum M_{pb}^*$). A typical “switch-case” programming structure is adopted to evaluate which design requirement is violated and an appropriate ameliorative measure is taken. If the shear or bending moment capacity is found to be less than their corresponding expected demands, the algorithm revises the beam design such that the new section size is one index lower than the old one in the section sub-database. If the SCWB criterion is not met, the column section is increased. Since a typical connection (with the exception of the roof) consists of two columns (one in the upper story and the other in the lower story), the one that is adjusted is determined based on the relationship between the plastic section modulus of those two columns. The column in the upper story is selected if its plastic section modulus is found to be 50% less than the one in the lower story. Otherwise, the column in the lower story is selected. Note that the assumed 50% threshold could be adjusted by the user. The entire process of evaluating the strength and SCWB ratio for the connections is repeated until all members and connections satisfy the design requirements.

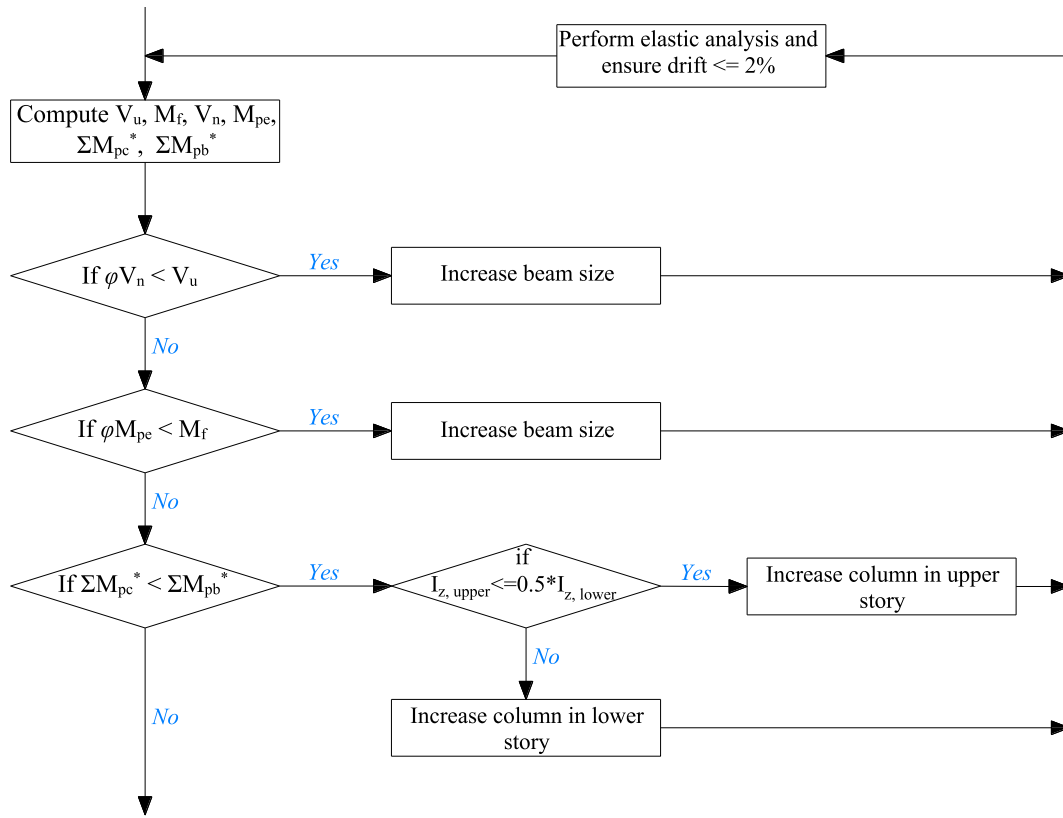
At this stage, the algorithm has generated SMRF member sizes that satisfy all design requirements. Subsequently, the member sizes are further adjusted to account for ease of construction. The beam sections are first adjusted such that identical section sizes are used over a

specified number of adjacent stories. The detailed algorithm for adjusting beam sizes is presented in Fig. 8. Starting from the roof and moving downward, the beams are grouped by the number of adjacent floors specified by the user. Then, the algorithm determines whether the beams at the current floor level i and the next lower level ($i-1$) belong to the same group (i.e., whether beams at levels i and $i-1$ are supposed to have identical section sizes). If they belong to the same group, the appropriate adjustments are made such that they have the same section size. Otherwise, the algorithm takes another set of different actions to ensure that the beam at level $i-1$ has a larger depth and moment of inertia than the one at level i . This process is coded using a compound “if-else” structure and, as shown in Fig. 8, there are ten possible cases in total. It should be noted that the algorithm shown in Fig. 8 is generic and can be used for other LFRS’s (e.g., RC moment frames) to ensure ease of construction. After adjusting the beam sizes, the size of those columns that do not meet the SCWB criterion is increased. The previously described process is repeated for columns to ensure that the same section size is used over a user-specified number of adjacent stories. The detailed design of the column splice is not implemented in the current version of the AutoSDA platform. Additionally, the response of the splice connection is not explicitly represented in the nonlinear structural model.

The algorithm eventually generates a design that complies with the relevant code and standards while accounting for ease of construction. An additional elastic analysis is performed to obtain the updated design information, including the column-to-beam flexural strength ratio, demand-to-capacity ratio, design drifts, etc.

3.4. Object-Oriented-Programming structure

The AutoSDA platform is written in Python using the object-oriented paradigm. The overall programming structure for the seismic design module including the associated class definitions are shown in Fig. 9. As noted earlier, the input parameters for this module are the building geometry (number of stories, number of bays, story heights, bay widths, and the number of SMRFs in each principal direction), loads (dead and live loads on each floor), site conditions (mapped spectral acceleration parameters and site class), and the parameters



Nomenclature:

- V_u : required shear strength based on probable moment and shear caused by gravity
 M_f : probable maximum moment at column face based on expected yield strength
 V_n : shear capacity of beam
 M_{pe} : plastic moment of the beam based on expected yield stress
 ΣM_{pc}^* : the sum of the moments in the column above and below the joint at the intersection of the beam and column centerlines
 ΣM_{pb}^* : the sum of the moments in the beams at the intersection of the beam and column centerlines
 $I_{z, upper}$: moment of inertia of column in upper story
 $I_{z, lower}$: moment of inertia of column in lower story

Fig. 7. Overview of the sub-algorithm used to ensure that the design requirements for all beam-column connections are satisfied.

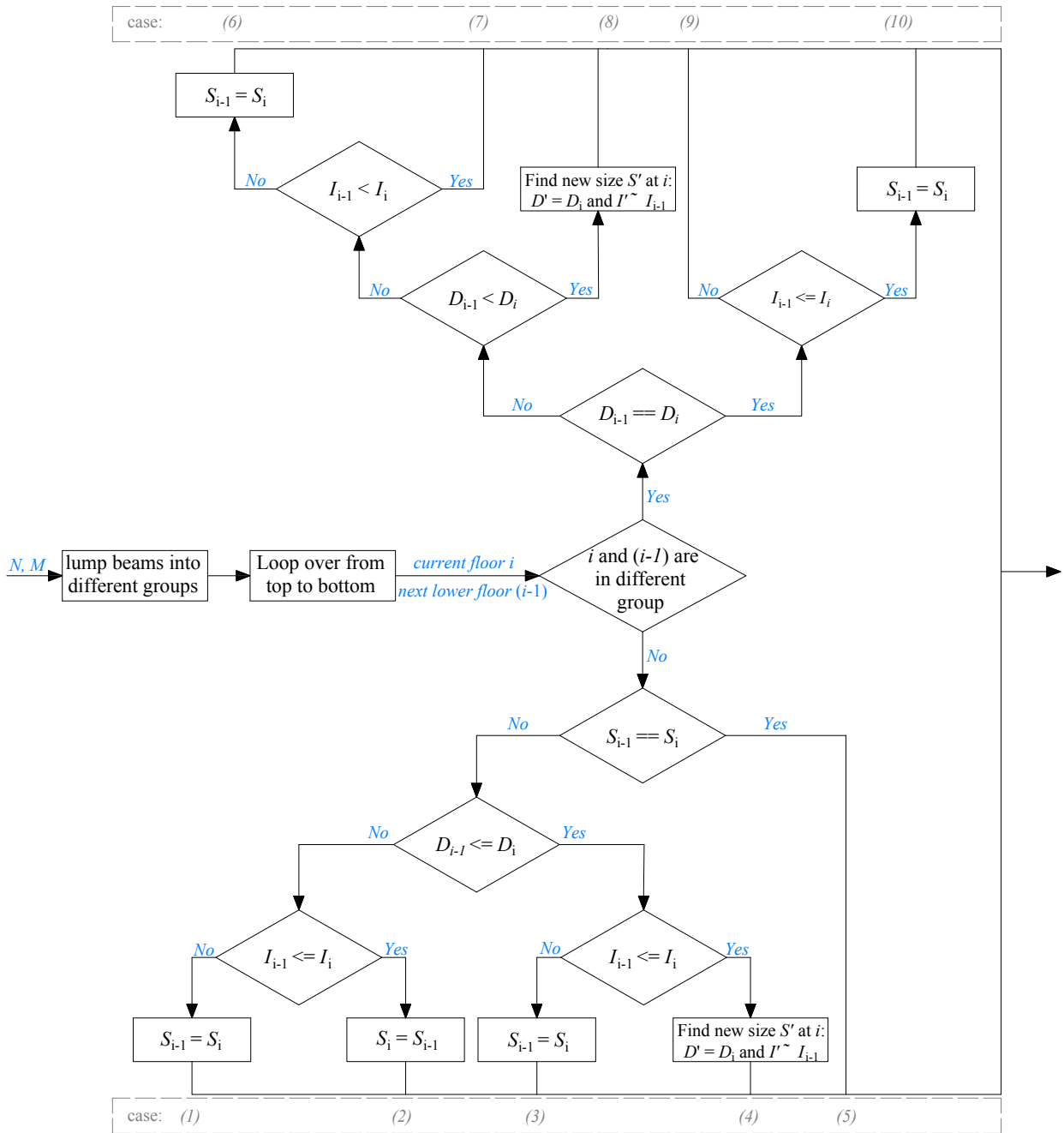
used in the equivalent lateral force procedure (response modification coefficient, importance factor, deflection amplification factor, redundancy factor, and parameters used to estimate fundamental period). Additional engineering-judgement based input parameters include the user-specified allowable drift limit, $I_{col,ext}/I_{col,int}$, $Z_{bm}/Z_{col,int}$, the threshold for the ratio of the column moment of inertia in upper story to the one in the lower story (I_{lower}/I_{upper}), the number of adjacent stories that are required to have the same section size, and a list of discrete values for the beam/column section depths that will be considered. The design module outputs the member sizes, design drifts, connection detailing, and demand-to-capacity ratios. The interface for the inputs and outputs are a set of .csv files.

The seismic design module consists of six classes: *Building*, *ElasticAnalysis*, *ElasticOutput*, *Column*, *Beam*, and *Connection*. The *Building* class reads all the building information from input files, computes the necessary “higher-level” variables used for seismic design, initializes the member sizes and adjusts the designs. The *ElasticAnalysis* class incorporates a series of member functions that generate a set of text files, which are used to construct the OpenSees model for the

elastic analysis. The role of the *ElasticOutput* class is to extract the member axial, shear, and flexural demands from the analysis output files, implement the necessary load combinations and determine the governing demands for the beams and columns. The *Beam*, *Column*, and *Connection* classes check whether each component satisfies the relevant requirements and stores the pertinent data (e.g., strengths, demands, and lateral bracing).

4. Nonlinear model construction and analysis

The nonlinear model construction and analysis (NMCA) module takes the design results as input, constructs nonlinear structural models, and performs static and/or dynamic structural response analyses. As noted earlier, the current version of the AutoSDA platform only allows the user to build a 2D structural model using the concentrated plasticity model for the beam-column elements. However, future versions can be adapted to incorporate 3D models and other types of beam-column elements (e.g., elements with fiber sections or finite length hinge). The overall programing structure and associated class definition for the



Nomenclature:

N : number of stories for the building

M : number of stories that should be adjusted to have identical size

S_i : section size at floor level i

D_i : section depth at floor level i

I_i : moment of inertia of section at floor level i

$S_i == S_{i-1}$: condition if current section size is the same as the lower section size

$S_i = S_{i-1}$: change the section size at floor level i to be the same as that at level ($i-1$)

Fig. 8. Overview of the sub-algorithm used to revise the beam sizes for ease of construction.

NMCA module is shown in Fig. 10. There is only one class: *Non-linearModelGeneration*. It provides a set of static methods to write the text files, which are further used to generate the nonlinear structural models in OpenSees and perform eigenvalue, static pushover over, and

dynamic analyses (including incremental dynamic analysis to collapse). Once all the necessary text files have been generated, the OpenSees execution file is called directly from the Python environment. Additionally, the NMCA module provides a set of independent functions to

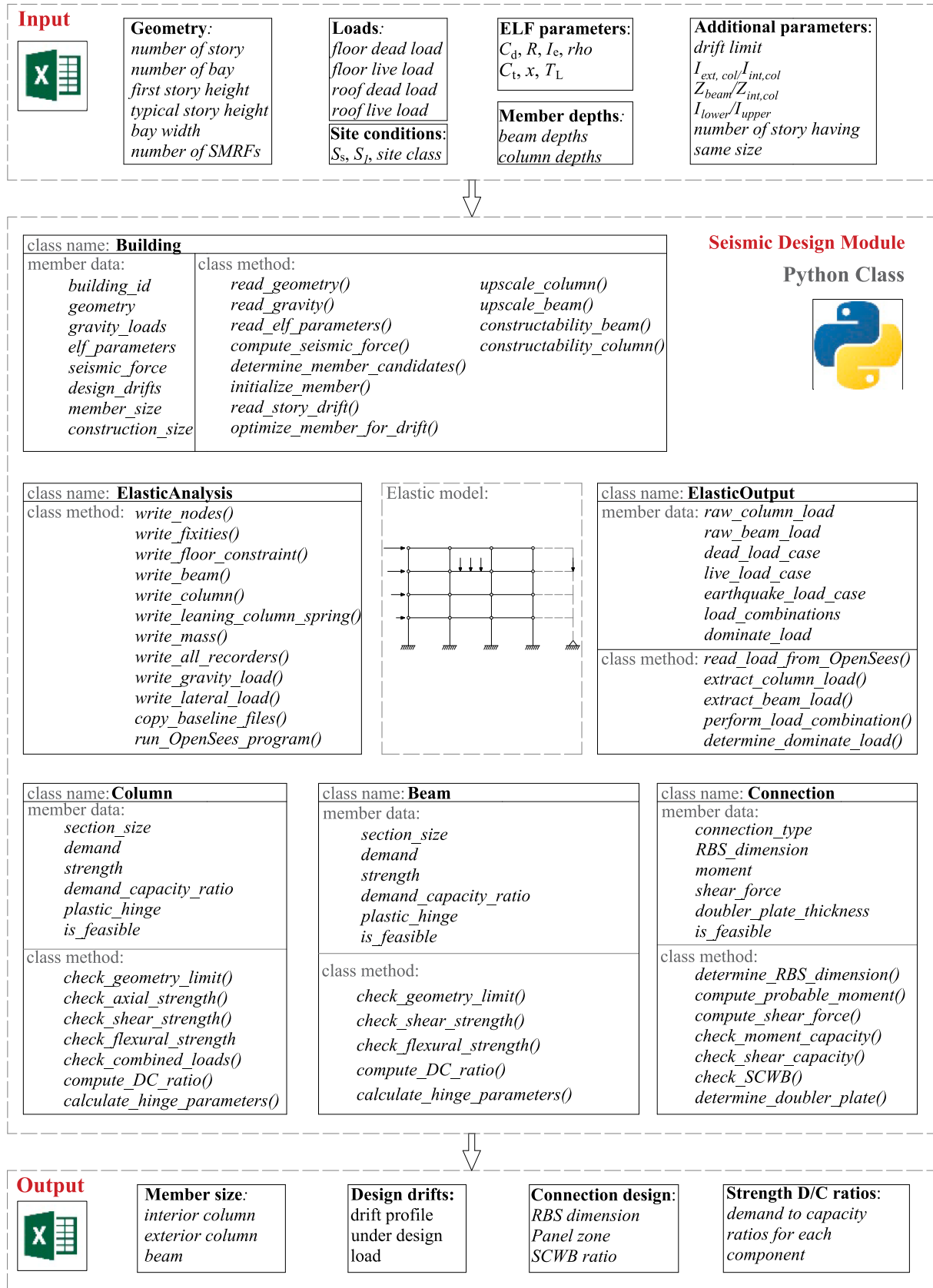


Fig. 9. Programing structure of the seismic design module.

visualize the analysis results. The building weight, nodal forces and nodal displacements from the pushover analysis is used to plot the normalized base shear vs. roof drift. Peak story drifts and floor

accelerations and residual story drifts can also be extracted from the dynamic analysis results. Three statistical methods, maximum likelihood estimation, minimizing sum of squared errors and *Probit*

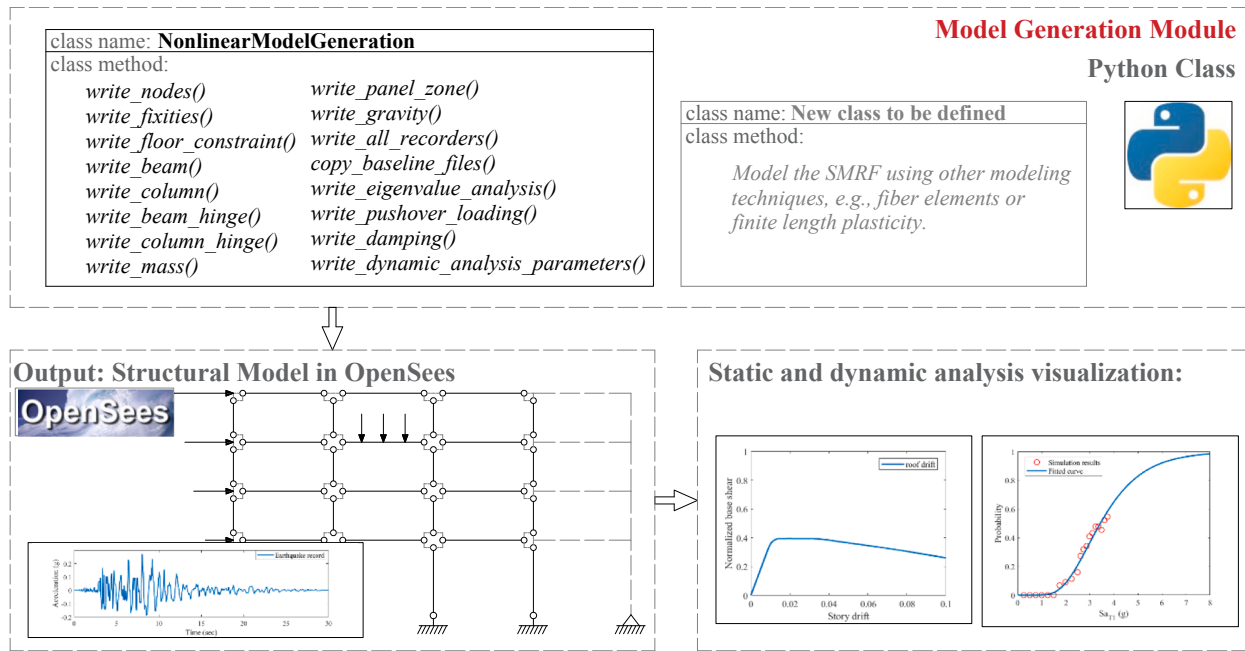


Fig. 10. Programming structure of the NMCA module.

regression, are available to fit the extracted engineering demand parameters at different intensities to a lognormal distribution [41], which can then be visualized (e.g., collapse fragility curve).

5. Illustrative example

5.1. Seismic design

Unlike currently available commercial software, which only provide the final results, the seismic design module of AutoSDA documents all iterations of the design process. In other words, the AutoSDA design process is not a black box and can be carefully scrutinized by the user. A five-story office building located in Los Angeles (34.008°N, 118.152°W) is used to illustrate how the design module iteratively updates member sizes to meet the relevant design criteria. As shown in Fig. 11, the building has SMRFs in the two principal directions. The floor and roof dead load are taken to be 80 psf (3.83 kN/m²) and 67.5 psf (3.23 kN/m²), respectively. The live loads for a typical floor and the roof are

taken as 50 psf (2.39 kN/m²) and 20 psf (0.96 kN/m²), respectively, in accordance with Table 4.3-1 of ASCE 7-16 [1]. The selected location corresponds to site class D and the mapped response spectral accelerations are $S_s = 2.25g$ and $S_1 = 0.6g$. Besides these variables, the following input parameters are defined: the range of member depths for the columns and beams are set as W14 and W21-W36, respectively. $I_{col,ext}/I_{col,int}$ and $Z_{bm}/Z_{col,int}$ are both set to 0.70. For ease of construction, it is assumed that the SMRF beam and column sizes are the same every two stories.

After receiving the input parameters, the program begins the iterative seismic design process. The variation in member sizes during a subset of the iterations is illustrated in Fig. 12. To begin, all beams and columns are assigned the maximum allowable size from their respective sub-database based on the user-specified depths (Fig. 12(a)). Then, an elastic analysis is performed, and it is determined that the minimum drift along the frame height is in the uppermost story. Consequently, the interior column size in the uppermost story is reduced and the size of the exterior columns and beams are determined based on the

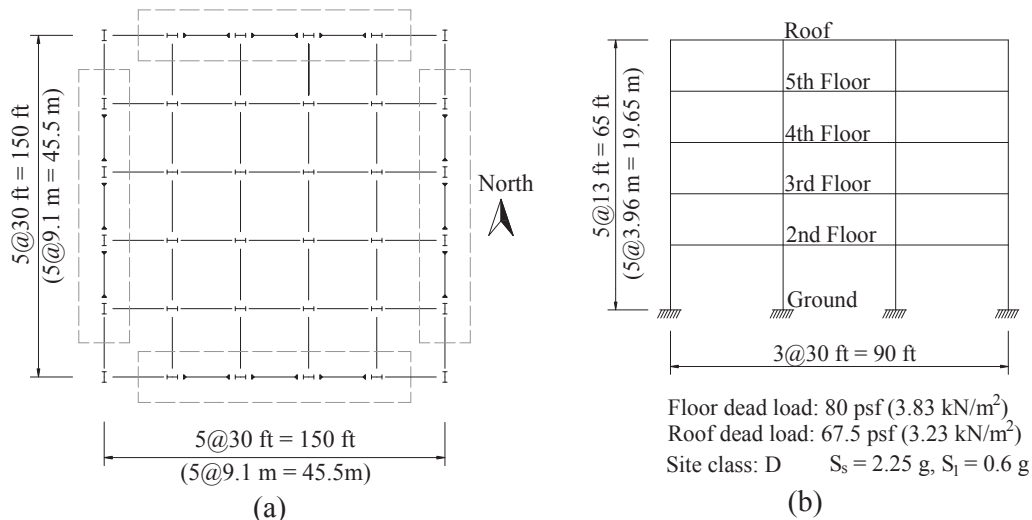


Fig. 11. Building case used to illustrate the AutoSDA design process: (a) floor plan and (b) elevation of SMRF.



Fig. 12. Changes in member sizes at different design stages: (a) initial sizes, (b) member sizes after first optimization for drift requirement, (c) most economical sections satisfying drift requirement, (d) section sizes after checking requirements for beams and columns, (e) design after checking strong-column-weak-beam criterion, (f) code-conforming design, (g) member sizes after adjusting beams for ease of construction, and (h) final design.

predefined $I_{col,ext}/I_{col,int}$ and $Z_{bm}/Z_{col,int}$ values (Fig. 12(b)). The elastic analysis is repeated with the new section sizes to determine which members should be optimized in the next step. After 54 iterations, the program reaches the point where the section sizes that have the smallest moment of inertia and meet the drift requirement are selected (Fig. 12(c)). Subsequently, those section sizes are checked against the requirements for beams and columns, but no revision is needed (Fig. 12(d)). Each connection is evaluated, and it is found that all connections meet the SCWB criterion. Therefore, no revision is made (Fig. 12(e)). After all design requirements are satisfied, the program outputs the code-conforming section sizes (Fig. 12(f)). The beam and column sizes (Fig. 12(g)) are adjusted to account for ease of construction, which produces the final design shown in Fig. 12(h).

5.2. Efficiency in time needed to complete design

The time needed to design a single building is used as a measure of the efficiency of the AutoSDA platform. To gain comprehensive insight into the design duration (from start to generation of the final design), 11 buildings with different numbers of stories and bays are designed using the AutoSDA platform on a computer with 16 GB RAM and Intel i7 3.6 GHz processor. The number of stories and bays for those 11 buildings range from 1 to 19 and 1 to 5, respectively. All other design information (gravity load, site condition, story heights, and plan dimensions) are the same as the building presented in Section 6.1 except that the bay width for those 11 buildings is 20 ft (6.1 m). The design duration for those buildings is listed in Table 1. It is worth noting that these design durations do not include the time needed to collect the

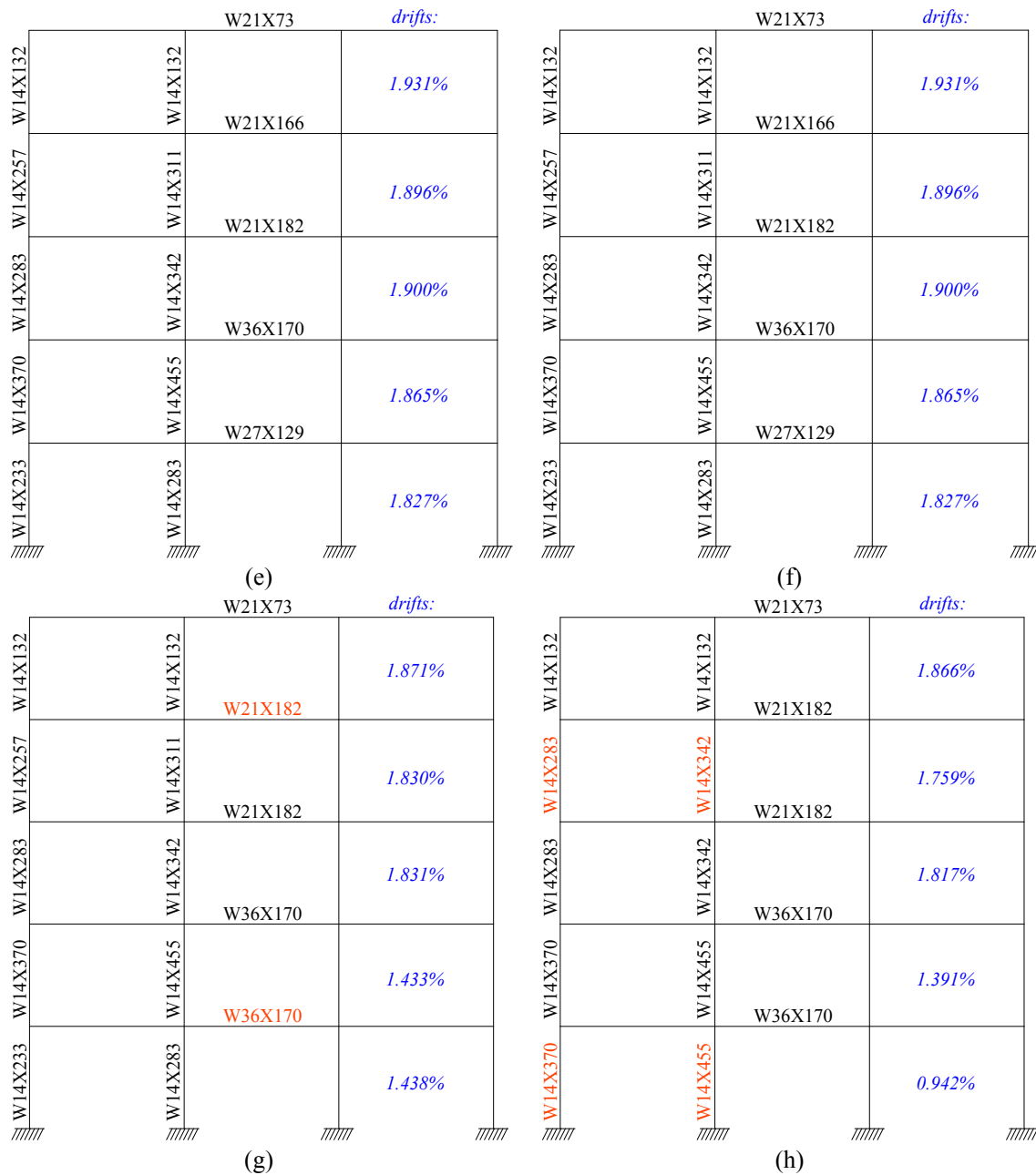


Fig. 12. (continued)

input data (including building geometry, loads, ELF parameters, site conditions, and member depths). All the input data are considered to be immediately available once the building information is given. The design process for low-rise, mid-rise, and high-rise buildings takes

approximately 1 min, 5 min, and 40 min, respectively. This is significantly less than the time it would take to perform these designs manually, which is an indication of the efficiency of the computational platform.

Table 1

Design duration for buildings with different numbers of stories and bays.

Number of stories Number of bays	Design Duration (seconds)				
	1-story	5-story	9-story	14-story	19-story
1-bay	21	\ ^a	\	\	\
3-bay	28	162	295	1128	1368
5-bay	31	161	295	1902	2478

^a The cells marked with “\” represent cases where either the design is impractical or no design solution exists for the combination of bays and the number of stories.

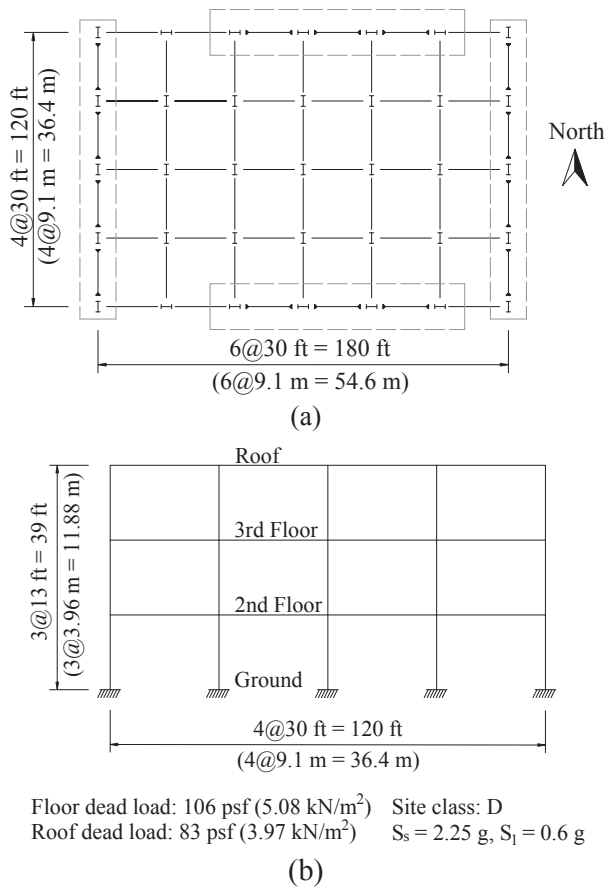


Fig. 13. Three-story building used in the ATC 123 project [23]: (a) floor plan and (b) elevation view.

5.3. Verification of the seismic design module

Despite its efficiency and the ability to keep track of the entire process, the reliability of the generated designs is a major concern. To address this issue, three buildings designed and reviewed by industry structural engineers and researchers are selected to assess the reliability of the module.

5.3.1. Comparison with designs produced by Englekirk structural engineers

As part of the ATC-123 project [23], three- and nine-story SMRF office buildings located in Los Angeles were designed by Englekirk Structural Engineers (<https://www.englekirk.com>). As shown in Figs. 13 and 14, both buildings have two SMRFs in the each orthogonal direction. The SMRF in the North-South direction is designed using AutoSDA and compared with the Englekirk designs. The dead loads on a typical floor and roof for both buildings are 106 psf (5.08 kN/m²) and 83 psf (3.97 kN/m²), respectively. The site class is D and the associated spectral response accelerations are $S_s = 2.25g$ and $S_1 = 0.6g$. The comparison between the Englekirk designs and the ones generated by AutoSDA are summarized in Tables 2 and 3. It is observed that the two designs are quite comparable. Fig. 15 shows a comparison of the design story drifts, which are also similar. Based on these two metrics, it is concluded that the designs generated by AutoSDA are comparable to the overall Englekirk designs (see Tables 2 and 3).

5.3.2. Comparing with designs produced by researchers

A four-story office building located in Los Angeles was developed by Lignos [3] and its seismic performance was evaluated. As shown in Fig. 16, the building has two SMRFs in each orthogonal direction. The SMRF in the North-South direction was designed using the AutoSDA

platform. The floor and roof weights are 1050 kips (4670 kN) and 1200 kips (5338 kN), respectively. The site class is D and the associated spectral acceleration parameters are $S_s = 1.5g$ and $S_1 = 0.6g$. The allowable drift limit is taken as 2.5% because the building was designed using the 2003 International Building Code [42]. Additional details of the building can be found in Chapter 5 of Lignos [3]. The comparison between the design reported by Lignos [3] and the one generated by AutoSDA is summarized in Table 4. It is observed that the section sizes for the two designs are quite comparable, which again demonstrates the reliability of the AutoSDA platform.

5.4. Comparing features of AutoSDA with commercial software: RAM and SAP 2000

The seismic design module is compared with commercial software to illustrate its advantages and limitations. *RAM Steel* and *SAP 2000*, which are commonly used structural design software, are selected for this purpose. The comparison is summarized in Table 5. In *RAM Steel*, the user manually constructs the elastic SMRF model using a graphical user interface (GUI). In contrast, this process is automated in the AutoSDA platform and *SAP 2000* without the user's intervention. As for the design process, *RAM Steel* can only evaluate the feasibility of a specified design. In other words, the member sizes must be assigned and adjusted by the user. While *SAP 2000* provides an "auto-list" function that implies seismic design automation, it simply determines the member sizes based on the demands from the previous elastic analysis without updating the demands. The user needs to perform a manual iteration to finalize the design. These issues are not present in the seismic design module of AutoSDA. As noted earlier, another advantage of AutoSDA is that it automatically tracks each change in member size during the entire design process. Both *RAM Steel* and *SAP 2000* do not have this capability. Moreover, the time spent to perform a design using *RAM Steel* and *SAP 2000* depends on the user and is on the order of several hours to days. In contrast, the longest running time for a single design using AutoSDA is less than one hour. A limitation of the AutoSDA platform is that its current version is limited to 2D modeling and it cannot deal with torsional irregularity or other three-dimensional effects such as biaxially loaded columns. However, given the adopted object-oriented programming structure and modular framework, AutoSDA can be easily extended to consider three-dimensional issues. Also, the current version of AutoSDA can only produce designs using equivalent lateral force analyses, whereas both *SAP2000* and *RAM Steel* can accommodate RSA- and RHA-based designs. Lastly, AutoSDA does not have a graphical user interface (GUI) which is included in both *SAP 2000* and *RAM Steel*.

5.5. Nonlinear static and dynamic analysis of SMRF buildings

The three-story building used in the ATC 123 project is used to illustrate the features of the NMCA module. Both nonlinear static and dynamic analyses, including incremental dynamic analyses, are performed. The pushover loading pattern is determined using the equivalent lateral force procedure prescribed in Chapter 12 of ASCE 7-16 [1] and assuming that the response is governed by the first mode of vibration. The pushover response for the North-South SMRF is shown in Fig. 17. The frame base shear is normalized with respect to its tributary seismic weight. The maximum base shear observed in Fig. 17 is 0.39, whereas the design base shear is 0.1. The overstrength factor, which is defined as the ratio of the maximum base shear to the code-design base shear [20], is computed as 3.86, which is greater than the code-specified minimum of 3.0. The period-based ductility ratio, which is defined as the ratio of the roof drift corresponding to a 20% drop in the maximum base shear to the yield roof drift [20], is also computed. The drift corresponding to a 20% drop in the maximum base shear is 0.075 and the yield drift is 0.011. Therefore, the ductility ratio is 6.97. The resulting values for the overstrength factor and ductility ratio serve as

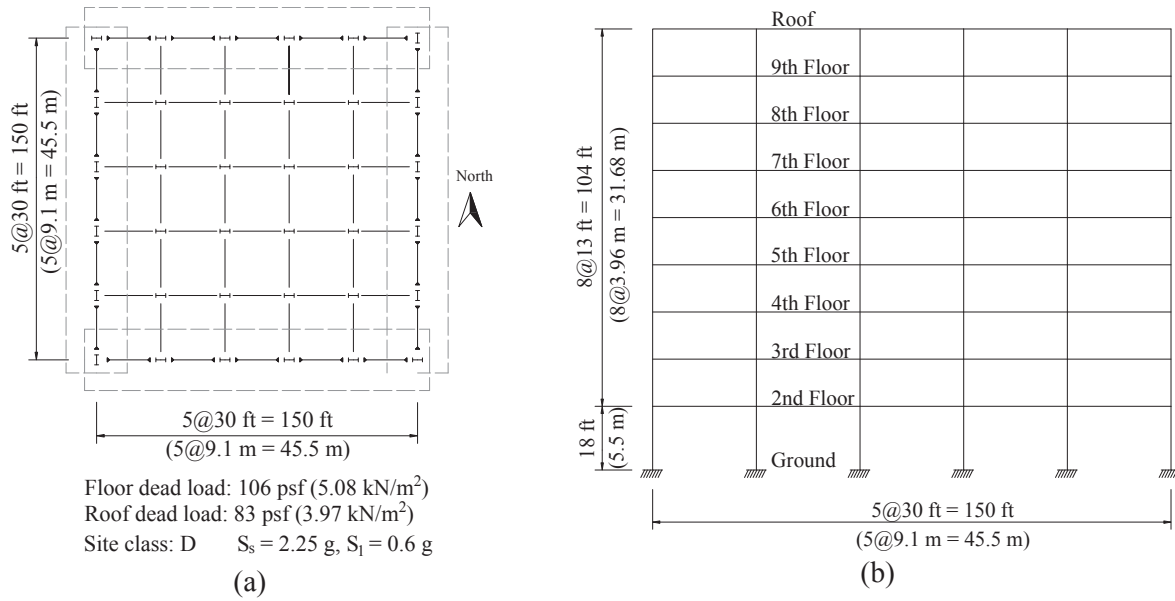


Fig. 14. Nine-story building used in the ATC 123 project [23]: (a) floor plan and (b) elevation view.

Table 2

Comparing member sizes between designs produced by *Englekirk* and the AutoSDA platform for the three-story building.

Story	Englekirk			Seismic Design Module		
	Exterior column	Interior column	Beam	Exterior column	Interior column	Beam
3	W14X211	W14X311	W27X94	W14X132	W14X159	W27X94
2	W14X311	W14X370	W33X130	W14X311	W14X455	W30X132
1	W14X311	W14X370	W33X130	W14X311	W14X455	W30X132

Table 3

Comparing member sizes between designs produced by *Englekirk* and the AutoSDA platform for the nine-story building.

Story	Englekirk			Seismic Design Module		
	Exterior column	Interior column	Beam	Exterior column	Interior column	Beam
9	W14X233	W14X311	W27X94	W14X132	W14X145	W33X130
8	W14X233	W14X311	W30X116	W14X311	W14X455	W33X130
7	W14X370	W14X398	W36X150	W14X311	W14X455	W33X130
6	W14X370	W14X398	W36X150	W14X398	W14X550	W33X152
5	W14X398	W14X426	W36X182	W14X398	W14X550	W33X152
4	W14X398	W14X426	W36X194	W14X426	W14X605	W36X160
3	W14X455	W14X500	W36X232	W14X426	W14X605	W36X160
2	W14X455	W14X500	W36X232	W14X500	W14X665	W36X170
1	W14X455	W14X550	W36X232	W14X500	W14X665	W36X170

further indication of the reliability of the AutoSDA-generated designs.

The seismic performance of the three-story building is assessed using truncated incremental dynamic analyses (IDA) [39]. The set of 44 (22 pairs) far-field ground motions selected as part of the FEMA P695 project [20] is used. The scaling for the truncated IDAs is performed such that the median spectral acceleration of the record-set matches the target intensity levels, which ranges from 20% to 300% of the spectral acceleration corresponding to the maximum considered earthquake (Sa_{MCE}).

The truncated results are used to generate a collapse fragility, where a lognormal distribution function is used to fit the simulation data via the maximum likelihood method. The collapse fragility curve for the three-story building is shown in Fig. 18. The collapse margin ratio

(CMR), which is defined as the ratio of the median collapse spectral acceleration to Sa_{MCE} , is computed. The median collapse capacity observed from Fig. 18 is 3.44 g and the Sa_{MCE} is 1.22 g, which corresponds to a CMR of 2.81. This result is further adjusted by applying a spectral shape factor of 1.36 (FEMA P695 Table 7–1). Thus, the adjusted collapse margin ratio (ACMR) is 3.82. According to Table 7–3 of FEMA P695, the minimum permissible ACMR, which is based on an MCE level collapse probability of 10%, is 1.83. The ACMR for the three-story building is twice the permissible value, indicating acceptable collapse performance. This observation serves as further evidence of the reliability of the AutoSDA-based designs.

5.6. Adaptability of the AutoSDA platform and possible future extensions

The AutoSDA platform makes extensive use of object composition such that all data and functions are encapsulated into abstract classes. The object-oriented programming structure allows future developers to augment the platform and enhance its functionality. For example, it could be easily adjusted to construct structural models using other types of nonlinear beam-column elements (instead of or in addition to the concentrated plasticity model) by simply adding a new class that can serve as an alternative to the current *NonlinearModelGeneration* class shown in Fig. 10. An “if-else” switch can then be added prior to the model generation module to determine the desired modeling technique. Moreover, the platform could also be extended to implement 3D structural modeling which can be used to investigate issues such as torsional irregularities and orthogonal loading effects.

In addition to being object-oriented, AutoSDA is highly modular. This allows future developers to conveniently extend the platform by incorporating other applications without changing the code in the existing modules. One possible extension is the addition of two more modules that incorporate ground motion selection and economic loss assessment. The former will take a target response spectrum and its standard deviation over a range of periods as input and will output a list of scaled or unscaled earthquake records from the PEER NGA-WEST2 [40] database. Those ground motion records can then be passed to the NMCA module for nonlinear response history analyses. The loss assessment module can take the engineering demand parameters from the NMCA module and use them to evaluate earthquake-induced financial losses using the FEMA P58 methodology [12]. The implementation of those two modules would produce an end-to-end computational

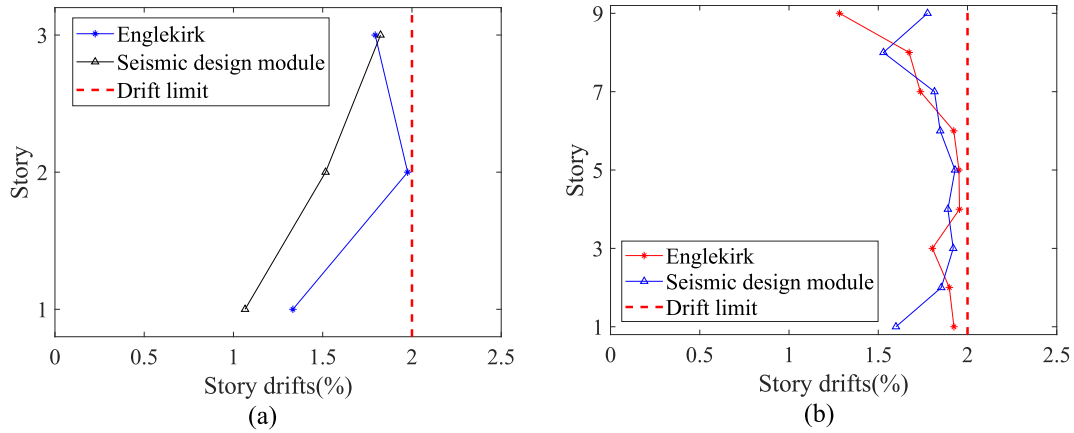


Fig. 15. Comparing design story drifts for the Englekirk and AutoSDA designs: (a) three-story and (b) nine-story buildings.

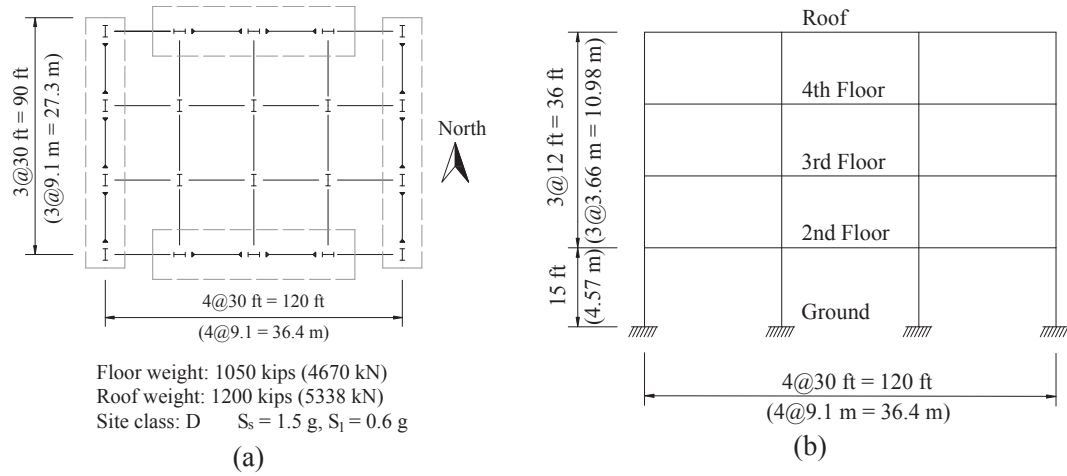


Fig. 16. Four-story office building reported by Lignos [3].

Table 4

Comparing member sizes between designs produced by the AutoSDA platform and Lignos [3].

Story	Seismic Design Module			
	Exterior/Interior column	Beam	Exterior/Interior column	Beam
4	W24X76	W21X93	W24X76	W21X68
3	W24X76	W21X93	W24X76	W21X68
2	W24X117	W27X102	W24X131	W27X102
1	W24X117	W27X102	W24X131	W27X102

platform for performance-based seismic design.

The entire platform is well-encapsulated such that it could incorporate parallel computing. As noted earlier in the Introduction, the number of buildings that are being used in archetype studies continues

Table 5

Comparing features of RAM Steel, SAP 2000, and AutoSDA.

Features	RAM Steel	SAP 2000	Seismic design module
Auto-modeling	×	✓	✓
Auto-iterative design	×	×	✓
Auto-tracking of the design process	×	×	✓
3D modeling (torsional irregularity)	✓	✓	×
Design duration	depends on the user	depends on the user	less than 1 h
RSA and RHA-based design	✓	✓	×
Graphical User Interface (GUI)	✓	✓	×

to increase (as high as on the order of 10^3). By calling the encapsulated package in parallel, multiple building cases can be simultaneously evaluated, which greatly improves the efficiency. Lastly, while the algorithms and class definitions in the current version of AutoSDA are developed for SMRFs, the workflow and structure can be adopted to incorporate other types of structural systems such as reinforced-concrete moment frames.

6. Conclusion

This paper presents a Python-based platform that automates the seismic design, nonlinear structural model generation, and response simulation of steel special moment resisting frames (SMRFs). The first module of the automatic seismic design and analysis (AutoSDA) platform takes building configuration, loads, and site parameters as input and outputs SMRF designs that comply with the latest building code

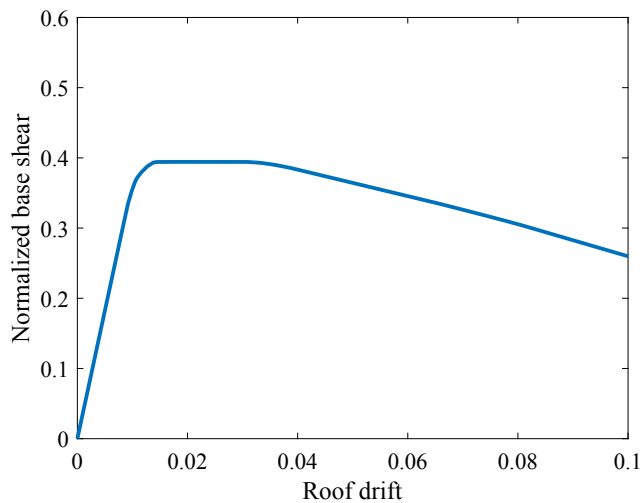


Fig. 17. Monotonic pushover curve for the three-story building.

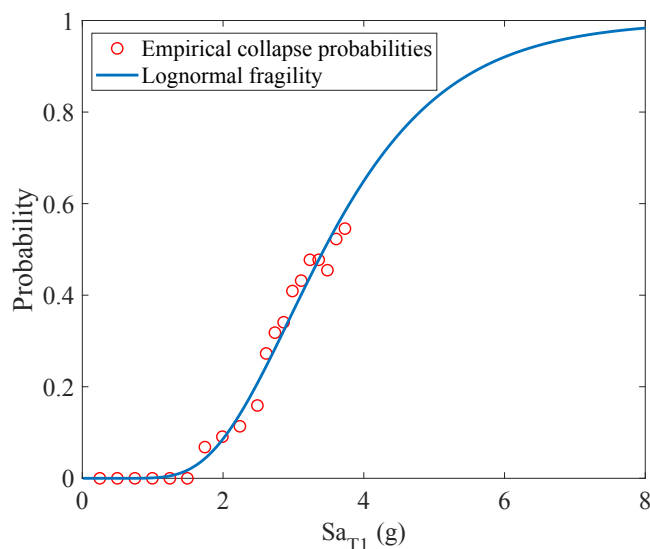


Fig. 18. Collapse fragility for the three-story building.

provisions while accounting for ease of construction. A second module constructs two-dimensional nonlinear structural models in OpenSees based on the generated designs and performs nonlinear static and dynamic analyses towards a comprehensive evaluation of seismic performance. The efficiency, reliability, and accuracy of AutoSDA are demonstrated using several illustrative examples. The modular framework and object orientated programming structure makes the platform easily adaptable. Potential future enhancements include the use of alternative strategies to account for beam-column material nonlinearity, 3D modeling and economic loss assessment. The broad implication of AutoSDA is a drastic reduction in the time and effort involved in performance-based seismic design. Moreover, it can be used to develop a database of archetype steel moment frame buildings towards the development of analytics-driven design methodologies. It is worth noting that the development details (e.g., platform structure and algorithm) documented in this paper can be used to create similar platforms for other types of structural systems. A key limitation of the current version of AutoSDA is that it only allows the design of SMRFs using the ELF method. This limitation can be addressed by adding a feature that generates designs using the results from response spectrum and/or response history analyses. Some other limitations include the lack of column splice and foundation design, and the inclusion of a graphical user interface, all of

which could be easily incorporated in future versions. The Python source code for AutoSDA will be made available through the National Science Foundation Natural Hazards Engineering Research Infrastructure (NHERI) SimCenter.

CRedit authorship contribution statement

Xingquan Guan: Methodology, Software, Investigation, Data curation, Writing - original draft. **Henry Burton:** Conceptualization, Methodology, Funding acquisition, Resources, Writing - review & editing, Supervision. **Thomas Sabol:** Conceptualization, Methodology, Validation, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] ASCE. 7–16. Minimum design loads for buildings and other structures. Reston, VA 2016.
- [2] Hamburger RO, Malley JO. Seismic design of steel special moment frames. NIST GCR 2009:09–917.
- [3] Lignos D. Sidesway collapse of deteriorating structural systems under seismic excitations Doctoral dissertation Stanford University; 2008.
- [4] Lignos DG, Krawinkler H. Deterioration modeling of steel components in support of collapse prediction of steel moment frames under earthquake loading. J Struct Eng 2010;137:1291–302.
- [5] Lignos DG, Krawinkler H. A steel database for component deterioration of tubular hollow square steel columns under varying axial load for collapse assessment of steel structures under earthquakes. Proc. 7th Int. Conf. on Urban Earthquake Engineering (7CUEE), Center for Urban Earthquake Engineering, Tokyo Institute of Technology Tokyo 2010.
- [6] Flores FX, Charney FA, Lopez-Garcia D. Influence of the gravity framing system on the collapse performance of special steel moment frames. J Constr Steel Res 2014;101:351–62.
- [7] Elkady A, Lignos DG. Effect of gravity framing on the overstrength and collapse capacity of steel frame buildings with perimeter special moment frames. Earthquake Eng Struct Dyn 2015;44:1289–307.
- [8] Dimopoulos AI, Karavasilis TL, Vasdravellis G, Uy B. Seismic design, modelling and assessment of self-centering steel frames using post-tensioned connections with web hourglass shape pins. Bull Earthq Eng 2013;11:1797–816. <https://doi.org/10.1007/s10518-013-9437-4>.
- [9] Dimopoulos AI, Tzimas AS, Karavasilis TL, Vamvatsikos D. Probabilistic economic seismic loss estimation in steel buildings using post-tensioned moment-resisting frames and viscous dampers. Earthquake Eng Struct Dyn 2016;45:1725–41. <https://doi.org/10.1002/eqe.2722>.
- [10] Fang C, Yam MCH, Lam ACC, Xie L. Cyclic performance of extended end-plate connections equipped with shape memory alloy bolts. J Constr Steel Res 2014;94:122–36. <https://doi.org/10.1016/j.jcsr.2013.11.008>.
- [11] Moradi S, Alam MS. Feasibility study of utilizing superelastic shape memory alloy plates in steel beam–column connections for improved seismic performance. J Intell Mater Syst Struct 2015;26:463–75.
- [12] FEMA. Seismic performance assessment of buildings. Redwood City, CA: Applied Technology Council; 2012.
- [13] Ramirez CM, Miranda E. Significance of residual drifts in building earthquake loss estimation. Earthquake Eng Struct Dyn 2012;41:1477–93. <https://doi.org/10.1002/eqe.2217>.
- [14] Tzimas A, Kamaris G, Karavasilis T, Galasso C. Collapse risk and residual drift performance of steel buildings using post-tensioned MRFs and viscous dampers in near-fault regions. Bull Earthq Eng 2016;14:1643–62.
- [15] Haselton CB, Deierlein GG. Assessing seismic collapse safety of modern reinforced concrete moment-frame buildings. vol. 67. 2007.
- [16] Liel AB, Deierlein GG. Assessing the collapse risk of California's existing reinforced concrete frame structures: metrics for seismic safety decisions. vol. 69. 2008.
- [17] Haselton CB, Liel AB, Deierlein GG, Dean BS, Chou JH. Seismic collapse safety of reinforced concrete buildings. I: Assessment of ductile moment frames. J Struct Eng 2010;137:481–91.
- [18] Liel AB, Haselton CB, Deierlein GG. Seismic collapse safety of reinforced concrete buildings. II: Comparative assessment of nonductile and ductile moment frames. J

- Struct Eng 2010;137:492–502.
- [19] Deierlein GG, Liel AB, Haselton CB, Kircher CA. ATC 63 methodology for evaluating seismic collapse safety of archetype buildings. *Structures Congress 2008: Crossing Borders* 2008;1–10.
- [20] FEMA. P695. Quantification of building seismic performance factors. Redwood City, CA: Applied Technology Council 2009.
- [21] Zareian F, Lignos D, Krawinkler H. Evaluation of seismic collapse performance of steel special moment resisting frames using FEMA P695 (ATC-63) methodology. *Structures Congress 2010;2010*:1275–86.
- [22] Morfidis K, Kostinakis K. Seismic parameters' combinations for the optimum prediction of the damage state of R/C buildings using neural networks. *Adv Eng Softw* 2017;106:1–16.
- [23] FEMA P-2012. Assessing seismic performance of buildings with configuration irregularities. Redwood City, CA: Applied Technology Council; 2018.
- [24] Khoshnoudian F, Mehrparvar B, et al. Evaluation of IBC equivalent lateral response procedure for seismic isolated structures. *Australian Structural Engineering Conference 2005*. Engineers Australia 2005;992.
- [25] Whittaker AS, Constantinou MC, Ramirez OM, Johnson MW, Chrysostomou CZ. Equivalent lateral force and modal analysis procedures of the 2000 NEHRP Provisions for buildings with damping systems. *Earthquake Spectra* 2003;19:959–80.
- [26] Xu L, Grierson DE. Computer-automated design of semirigid steel frameworks. *J Struct Eng* 1993;119:1740–60.
- [27] Moharrami H, Grierson DE. Computer-automated design of reinforced concrete frameworks. *J Struct Eng* 1993;119:2036–58.
- [28] Kavlie D, Moe J. Automated design of frame structures. *Journal of the Structural Division* 1971;97:33–62.
- [29] McKenna F, Fenves GL, Scott MH. Open System for Earthquake Engineering Simulation. CA, USA: University of California Berkeley; 2000 <http://opensees.berkeley.edu>.
- [30] Guan X, Burton H, Shokrabadi M. A Database of Seismic Designs, Nonlinear Models, and Seismic Responses for Steel Moment Resisting Frame Buildings. *DesignSafe-CI* 2020. <https://doi.org/10.17603/ds2-8yc7-1285>.
- [31] American Institute of Steel Construction (AISC). Seismic provisions for structural steel buildings. ANSI/AISC 341-16. American Institute of Steel Construction: Chicago, IL 2016.
- [32] American Institute of Steel Construction (AISC). Specification for structural steel buildings. ANSI/AISC 360-16. American Institute of Steel Construction: Chicago, IL 2016.
- [33] Ibarra LF, Krawinkler H. Global collapse of frame structures under seismic excitations. CA: Pacific Earthquake Engineering Research Center Berkeley; 2005.
- [34] Ibarra LF, Medina RA, Krawinkler H. Hysteretic models that incorporate strength and stiffness deterioration. *Earthquake Eng Struct Dyn* 2005;34:1489–511.
- [35] Lignos DG, Hartloper AR, Elkady A, Deierlein GG, Hamburger R. Proposed updates to the ASCE 41 nonlinear modeling parameters for wide-flange steel columns in support of performance-based seismic engineering. *J Struct Eng* 2019;145:04019083.
- [36] Hwang S-H, Lignos DG. Earthquake-induced loss assessment of steel frame buildings with special moment frames designed in highly seismic regions. *Earthquake Eng Struct Dyn* 2017;46:2141–62.
- [37] Krawinkler H. Shear in beam-column joints in seismic design of steel frames. *Engineering Journal* 1978;15.
- [38] American Institute of Steel Construction (AISC). Prequalified connections for special and intermediate steel moment frames for seismic applications. ANSI/AISC 358-16. American Institute of Steel Construction: Chicago, IL 2016.
- [39] Vamvatsikos D, Cornell CA. Incremental dynamic analysis. *Earthquake Eng Struct Dyn* 2002;31:491–514.
- [40] Ancheta TD, Darragh RB, Stewart JP, Seyhan E, Silva WJ, Chiou BS, et al. *Peer NGA-West2 database* 2013.
- [41] Baker JW. Efficient analytical fragility function fitting using dynamic structural analysis. *Earthquake Spectra* 2015;31:579–99.
- [42] IBC. International building code. International Code Council; 2003.